25900671

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Value Addition Course(VAC) - MG3VACTXLA03 - TEXTILE PROCESSING AND FINISHING

(2024 ADMISSION ONWARDS)

ANSWER KEY

Duration: 1 Hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)

and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Which of the following materials are commonly used as flock fibers? [K] / [CO1]

a). Paper b). Rubber

c). Glass d). Nylon or Rayon

A: OPTION - D

2. In skein dyeing, yarn is dyed in the form of what? [U] / [CO3]

a). cones b). Hanks

c). Warp beams **d). Fabrics**

A: OPTION -B

3. Yarn dyeing is performed at which stage of textile production? [K] / [CO3]

a). Before the fibre is spun into yarn

b). After the yarn is spun but before weaving/knitting

c). After the fabric is woven d). Only on wool yarns

A: OPTION -B

4. What is the main purpose of tentering in textile finishing? [K] / [CO1]

a). To dye the fabric b). To remove lint

c). To dry and set the fabric width d). To add luster

A: POTION C

5. Duplex printing is especially useful for [U] / [CO4]

a). Socks and gloves b). Curtains and hangings

c). Bedsheets d). Carpets

A: OPTION C

6. The main purpose of a durable press finish is to [U] / [CO2]

a). Improve water resistance b). Reduce wrinkling and retain shape

c). Increase fabric weight d). Prevent color fading

A: OPTION B

7. Pad Dyeing is commonly used for. [U] / [CO3]

a). Loose fiber dyeing b). Yarn dyeing

c). Woven and knitted fabric dyeing d). Stock solution dyeing

A: OPTION C

8. The process of producing a water- marked (wavy) effect on fabric by calendaring is called. [K] / [CO1]

a). Moire b). Embossing

c). Glazing d). Friction finish

A: OPTION A

9. Which type of Dyeing method is used in Continuous dyeing of fabric? [K] / [CO3]

a). Jet dyeing b). Pad dyeing

c). Skein dyeing d). Dope dyeing

A: OPTION B

10. Rotary-cylinder singeing is especially suitable for. [U] / [CO1]

a). Denim fabrics b). Knitted fabrics

c). Velvet and pile fabrics d). Polyester blends

[1x10 = 10]

A: OPTION C

Part B

Fill in the blanks

Answer all questions

Each question carries **1** marks

11. Enhanced finishing is applied to fabric to improve their _____ properties, such as water resistance or wrinkle resistance. [U] / [CO2]
A: FUNCTIONAL

12. A defect caused by gritty particles in the color paste is called _____. [U] / [CO4]
A: SCUMMING

13. In continuous dyeing fabric passes through a dye bath and then through ……..rolllers. [U] / [CO3]
A: HEATED / PLAIN

14. The chemical used in Mercerization is _____. [K] / [CO1]
A: Sodium Hydroxide (NaOH)/ CASTIC SODA

15. Fibre dyeing is ideal for blended or ……..coloured effect [U] / [CO3]
A: HEATHERED/ GOOD / EFFECTIVE

16. The process of applying a protective adhesive coating to warp yarns before weaving is called _____. [K] / [CO1]
A: SIZING

17. A drawback of garment dyeing is poor _____ in thick areas. [U] / [CO3]
A: PENETRATION /COLOR EFFECT

18. In tumble dyeing, _____ ensures even penetration of dye. [U] / [CO3]
A:AGITATION/ MOVEMENT OF DYE OR FABRIC

19. In Bleaching,_____ are added to prevent rapid decomposition of bleaching agents. [U] / [CO1]
A: STABILIZER / CHEMICAL/AGENT

20. In dope dyeing ,colour pigments are added to the…...before the fibre extruded. [U] / [CO3]

[1x10 = 10]

A: POLYMER/ CHEMICAL

Part C

One word

Answer all questions

Each question carries **1** marks

21. Name one chemical used in shrinkage resistant finishes for wool. [K] / [CO2]
A: Chlorine/Resins/Hypochlorite/Hercosett resin

22. Which blanket carries the fabric during rotary printing? [K] / [CO4]
A: ENGRAVED ROLLER / ROLLERS

23. What is the process of removing sizing agents from the fabric? [K] / [CO1]
A: DESIZING

24. How many rollers are used to wind the fabric in jig dyeing? [U] / [CO3]
A: TWO

25. Which natural dye has antimicrobial properties? [K] / [CO3]
A: TURMERIC / PLANT

26. Name the type of finish that allows breathability but resists wetting. [K] / [CO2]
A: WATER PROOF / WATER REPELLENT

27. Name the Dyeing method used for Polyester-cotton blends to get different colors. [K] / [CO3]
A:CROSS DYE

28. Which printing method uses carved wooden blocks? [U] / [CO4]
A: BLOCK PRINTING

29. Which finish is essential for firefighter uniforms? [K] / [CO2]
A:FLAME RESISTANCE

30. Which synthetic dye is best for polyester fabrics? [K] / [CO3]

A: DISPERSE

[1x10 = 10]

Part D

Short Answer

Answer any **5** questions

Each question carries **2** marks

31. Which fabrics are commonly finished using napping and why? [K] / [CO1]
A: VELVET, **Napping raises the** fiber ends on the surface**, making the fabric** soft, warm, and fuzzy**, improving insulation and comfort.**

32. State three advantages of screen printing. [U] / [CO4]
A: bright and vivid designs**,** printing on a variety of fabrics**,** multi-colored patterns.

33. What is the shearing process in textiles? [U] / [CO1]
A: Shearing is a finishing process **where the** surface fibers of the fabric are trimmed **to a uniform height.**

34. Why is Winch Dyeing preferred for knitted fabrics? [K] / [CO3]
A: Ensures uniform dyeing **without damaging. Winch dyeing is a batch dyeing method where fabric is dyed in rope form.**

35. Discuss the uses and advantages of Stock dyeing? [U] / [CO3]

A: Produces color-fast yarns**.Allows creation of** heathered and blended effects**.**
Stock dyeing is a method of fiber dyeing where loose fibers, particularly wool and cotton, are dyed before spinning. It allows uniform coloration and is commonly used in producing tweeds, heathered fabrics, and mélange yarns.

36. Explain the process of Napping in textiles. [U] / [CO1]
[2x5 = 10]

A: **Fabric passes over rollers covered with fine wires or teasels.Wires lift fibers to create a nap (soft pile layer).Multiple passes may be required to raise sufficient nap.**

Part E

Short Essay

Answer any **2** questions

Each question carries **5** marks

37. Write a short note on synthetic dyes in textiles.

A: • **Synthetic dyes are man-made dyes produced from chemical compounds, unlike natural dyes.**

• **They are classified based on fiber type and application method, e.g., reactive, vat, direct, acid, disperse dyes.**

• **Advantages:**

1. **Wide range of vivid colors.** [U] / [CO3]
2. **High color fastness to washing, light, and rubbing.**
3. **Can be tailored for specific fibers (cotton, wool, polyester).**

• **Uses: Textile dyeing, printing, and finishing of natural and synthetic fibers.**

• **Importance: They allow consistent, reproducible colors in large-scale textile production**

38. Write a short note on Sizing and its importance in textile finishing.

A: • **Sizing is the process of applying a protective adhesive coating on warp yarns before weaving.**

• **Common sizing agents include starch, PVA, CMC, and synthetic resins.** [K] / [CO1]

• **Importance:**

1. **Increases strength and abrasion resistance of warp yarns during weaving.**
2. **Reduces yarn breakage and improves weaving efficiency.**

3. **Enhances smoothness and uniformity of the fabric.**

- **Outcome: Higher productivity in weaving and better quality fabric.**

39. Discuss the advantages and limitations of Beam Dyeing.

A: • **Beam Dyeing: Fabric wound on a perforated beam is passed through a dye bath under pressure.**

- **Advantages:**

1. **Suitable for delicate and lightweight fabrics.**
2. **Ensures uniform dyeing with minimal tension.**
3. **Can dye large quantities of fabric continuously.** [K] / [CO3]
4. **Reduces fabric distortion and creasing.**

- **Limitations:**

1. **Requires special machinery (beam dyeing machines).**
2. **Not suitable for very heavy or bulky fabrics.**
3. **Higher initial investment cost.**

40. Explain why discharge printing is more effective on dark backgrounds.

A: • **Discharge printing: A process where bleaching agents remove color from dyed fabric in the printed area.**

- **Effectiveness on dark fabrics:**

1. **Dark fabrics have more concentrated dye, so removing dye produces high contrast designs.**
2. **Light or neutral prints appear more vivid and prominent against a dark background.**
3. **Allows bright and sharp patterns without adding extra colors.**

- **Applications: Used for dark cotton, polyester-cotton blends, and printed home textiles.**

[U] / [CO4]
[5x2 = 10]

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